In the following code snippet, a model is trained on data_part1.csv. The other data chunks or fragmented data will be given to other organisations in the same format to ensure privacy of the entire dataset.

```
import pandas as pd
from sklearn.model_selection import
train_test_split
from sklearn.ensemble import
RandomForestClassifier
import joblib

# Load the dataset
data = pd.read_csv("data_part1.csv")

# Split the dataset into features and
target
X = data.drop(columns=['PatientID',
'Disease'])
y = data['Disease']

# Split data into train and test sets
X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2,
random_state=42)

# Create and train the Random Forest
classifier
rf_classifier = RandomForestClassifier(n_
estimators=40, random_state=42)
rf_classifier.fit(X_train, y_train)

# Evaluate the model
accuracy = rf_classifier.score(X_test,
y_test)
print("Accuracy:", accuracy)

# Save the model to a .pkl file
joblib.dump(rf_classifier, 'random_
forest_model_part1.pkl')
```

Using majority voting for merging multiple trained models

During the merging process, these models are aggregated such that they contribute equally to the final decision. The majority voting system functions by evaluating the predictions from each model on a given input; the final output prediction is determined by the most common result (majority vote) among all models. This democratic approach not only mitigates biases that might occur due to anomalies in individual datasets but also harnesses the diverse data characteristics learned across various environments, potentially leading to a more robust and generalised model. This method proves particularly effective in scenarios where no single model can be trusted to independently make accurate predictions, thereby enhancing the reliability and overall performance of the federated learning system.

In the following code snippet, multiple PKL files are used for getting results from multiple models. This is an example of the majority voting approach.

```
import pandas as pd
import joblib

# Load the test data
test_data = pd.DataFrame({
    'Age': [55],
    'Diabetic': [1],
    'BloodPressureLevel': [140]  #
Example blood pressure level
})

# Load the trained Random Forest models
model_part1 = joblib.load('random_
forest_model_part1.pkl')
model_part2 = joblib.load('random_
forest_model_part2.pkl')

# Predict disease using the first model
prediction_part1 = model_part1.
predict(test_data)

# Predict disease using the second
model
prediction_part2 = model_part2.
predict(test_data)

# Combine predictions using majority
voting
prediction_combined = 1 if (prediction_
part1.sum() + prediction_part2.sum())
>= 1 else 0

print("Combined Prediction:",
prediction_combined)
```

Federated learning offers a promising avenue for collaborative, privacy-preserving data analysis and model training across multiple decentralised entities. This technique not only allays concerns related to privacy and data sovereignty but also significantly reduces the bandwidth required for transmitting large datasets.

The research in federated learning is primarily focusing on improving algorithm efficiency, data security, and system scalability. Key research areas include the development of robust algorithms that can efficiently handle non-IID (independently and identically distributed) data across various nodes. Such algorithms strive to overcome the skewed data distribution that is characteristic of many real-world scenarios, where data collected by different nodes may vary significantly in size, type, and quality.

Federated learning presents a robust framework for handling data across dispersed networks while adhering to privacy and security requirements. As industries and institutions increasingly prioritise data confidentiality alongside the need for powerful analytical tools, federated learning is poised to become more pivotal. This progressive field promises to enhance the efficacy and efficiency of machine learning models across various applications and pioneer new frontiers in the ethical utilisation of Big Data. END

By: Dr Gaurav Kumar

The author is associated with various academic and research institutes for delivering expert lectures and conducting technical workshops on the latest technologies and tools.